



$$I_n f(x) = \int_a^b K_n(x, t) f(t) dt,$$

$$K_n(x, t) = \frac{1}{n!} t^n (x - t)_+^{n-1}$$

$$L_n f = \frac{1}{(n+1)!} f^{(n+1)}(\xi_n) L_n(x),$$

$$\int_a^b w(x) f(x) dx = \sum_{j=0}^n A_j f_j + R_n(f),$$

Hermite Interpolation
double nodes $H_n f(x) = \sum_{i=0}^m [h_{0i}(x)f(x_i) + h_{1i}(x)f'(x_i)]$ (1.3)
 $h_{0i}(x) = [1 - 2\ell_i(x)]\ell_i(x)$ (1.4)
 $h_{1i}(x) = (x - x_i)\ell_i(x)$, $i = 0, \dots, m$.
Newton's divided difference form
 $H_n(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)^2 + \dots + f[x_0, x_1, \dots, x_m](x - x_0)^m \dots (x - x_{m-1})$ (1.6)
 $H_n(x) = f(x) - H_n f(x) = \frac{f^{(n+1)}(\xi_n)}{(n+1)!} \prod_{i=0}^n (x - x_i)$, $\xi_n \in (a, b)$ (1.8)
 $h_{0i}(x) = \frac{(x - x_0)^2}{2!} \left[\frac{1}{(x - x_1)} - \frac{1}{(x - x_2)} \right] \dots \left[\frac{1}{(x - x_{i-1})} - \frac{1}{(x - x_{i+1})} \right] h_{1i}(x)$ (1.13)
 $N_n f(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)^2 + \dots + f[x_0, x_1, \dots, x_m](x - x_0)^m \dots (x - x_{m-1})$ (1.14)
 $R_n(x) = \frac{w(x)}{(n+1)!} f^{(n+1)}(\xi_n)$, $\xi_n \in (a, b)$.

- 1) Let $f \in C^3[0, 1]$.
- a) (2 points) Find $(P^1 f)(x)$, the polynomial interpolating the function f at the double node $x_0 = 1$ and the single node $x_1 = 0$.
- b) (3 points) Integrate $\int_0^1 (P^1 f)(x) dx$ to obtain a quadrature formula for $\int_0^1 f(x) dx$. Find its degree of precision.
- c) (2.5 points) Use Peano's theorem to find the remainder $R_1(f)$ of the quadrature formula in b).

$x_0 = 0$ double node $\Rightarrow$ stim $f(0), f'(0)$ $y_0 = \{0, 1\}$
 $x_1 = 1$ single node $\Rightarrow$ stim $f(1)$ $y_1 = \{0\}$
 $n = 1 + 1 + 1 - 1 = 2$ = nr. de informații pe care le avem
 $\{f(0), f'(0), f(1)\} - 1$

z	$f(z)$	$\Delta^1 f$	$\Delta^2 f$
$z_0 = x_0$	$f(0)$	$\frac{f(0) - f(0)}{x_0 - x_0} = f'(0)$	$\frac{f(1) - f(0) - f'(0)}{x_1 - x_0} = \frac{f(1) - f(0) - f'(0)}{1}$
$z_1 = x_0$	$f(0)$	$\frac{f(1) - f(0)}{x_1 - x_0} = \frac{f(1) - f(0)}{1}$	
$z_2 = x_1$	$f(1)$		

Newton's divided difference form

$$N_n(x) = f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)^2 + \dots + f[x_0, x_1, \dots, x_m](x - x_0)^m \dots (x - x_{m-1}) \quad (1.6)$$

$$N_2(x) = f(0) + f'(0)(x - x_0) + (f(1) - f(0) - f'(0))(x - x_0)^2$$

$$= f(0) + f'(0)x + (f(1) - f(0) - f'(0))x^2$$

$$b) \int_0^1 f(0) + f'(0)x + (f(1) - f(0) - f'(0))x^2 dx =$$

$$f(0) \cdot x \Big|_0^1 + f'(0) \frac{x^2}{2} \Big|_0^1 + (f(1) - f(0) - f'(0)) \frac{x^3}{3} \Big|_0^1$$

$$= f(0)(1 - 0) + f'(0)\left(\frac{1}{2} - 0\right) + (f(1) - f(0) - f'(0))\left(\frac{1}{3} - 0\right)$$

$$= f(0) + \frac{1}{2}f'(0) + \frac{1}{3}(f(1) - f(0) - f'(0))$$

$$= \frac{1}{3}f(0) - \frac{f(0)}{3} + \frac{f'(0)}{2} - \frac{f'(0)}{3} + \frac{f(1)}{3} = \frac{2f(0)}{3} + \frac{f'(0)}{6} + \frac{f(1)}{3}$$

$$= \frac{2f(0) + f(1)}{3} + \frac{f'(0)}{6} = \frac{4f(0) + 2f(1) + f'(0)}{6}$$

$$|n, 1| = \int_0^1 f(x) dx - \int_0^1 (P f)(x) dx \quad | \quad \int_0^1 (P f)(x) dx + R(f)$$

$$R(f) = \int_0^1 f(x) dx - \int_0^1 (Pf)(x) dx$$

$$\Rightarrow R(f) = \int_0^1 f(x) dx - \frac{4f(0) + 2f(1) + f'(0)}{6}$$

$$f(x) = e_0(x) = x^0$$

$$R(e_0(x)) = \int_0^1 x^0 dx - \frac{4e_0(0) + 2e_0(1) + e_0'(0)}{6}$$

$$= x \Big|_0^1 - \frac{4+2+0}{6} = 1 - 0 - 1 = 0$$

$$R(e_1(x)) = \int_0^1 x^1 dx - \frac{4e_1(0) + 2e_1(1) + e_1'(0)}{6}$$

$$= \frac{x^2}{2} \Big|_0^1 - \frac{0+2+1}{6} = \frac{1}{2} - 0 - \frac{3}{6} = \frac{1}{2} - \frac{1}{2} = 0$$

$$R(e_2(x)) = \int_0^1 x^2 dx - \frac{4e_2(0) + 2e_2(1) + e_2'(0)}{6}$$

$$= \frac{x^3}{3} \Big|_0^1 - \frac{0+2-0}{6} = \frac{1}{3} - \frac{2}{6} = 0$$

$$R(e_3(x)) = \int_0^1 x^3 dx - \frac{4e_3(0) + 2e_3(1) + e_3'(0)}{6}$$

$$\frac{x^4}{4} \Big|_0^1 - \frac{0+2-0}{6} = \frac{1}{4} - \frac{2}{6} = \frac{1}{4} - \frac{1}{3} \neq 0$$

$$= \frac{3-4}{12} = -\frac{1}{12}$$

degree of precision = 2

$$c) f(x) = H_2(f(x)) + R(f(x)) \Rightarrow R(f(x)) = f(x) - H_2(f(x))$$

$$I_n f(x) = \int_a^b K_n(x,t) f^{(n+1)}(t) dt$$

$\Rightarrow$ using Peano's Theorem for our test function $\Rightarrow$

$$K_n(x,t) = \frac{1}{n!} L((x-t)_+^n)$$

$$L f = \frac{1}{(n+1)!} f^{(n+1)}(\xi) L e_{n+1}$$

$$\int_a^b w(x) f(x) dx = \sum_{j=0}^m \lambda_j F_j(f) + R(f)$$

$$\Rightarrow R(f(x)) = \int_0^1 K_2(x,t) f'''(t) dt$$

$$K_2(x,t) = \frac{1}{2!} R((x-t)_+^2)$$

$$H_2(x) = f(0) + f'(0)x + \frac{(f(1) - f(0) - f'(0))}{2} x^2$$

$$= \frac{1}{2} (x-t)^2 \Big|_{t=0}^{t=1} \cdot x +$$

$$\int_0^1 (Pf)(x) dx + R(f)$$

quadrature formula

$$\begin{aligned}
R((x-t)_+^2) &= (x-t)_+^2 - \underbrace{(0-t)_+^2}_{=0} + \frac{\partial}{\partial x} (x-t)_+^2 \Big|_{x=0} \cdot x + \\
&+ \left(\underbrace{(1-t)_+^2}_{=0} - \underbrace{(0-t)_+^2}_{=0} - \frac{\partial}{\partial x} (x-t)_+^2 \Big|_{x=0} \right) x^2 \\
&= (x-t)_+^2 + \left(2(x-t)_+ \Big|_{x=0} \right) \cdot x + \left((1-t)^2 - 0 - 2(x-t)_+ \Big|_{x=0} \right) x^2 \\
&= (x-t)_+^2 + \left(2 \underbrace{(0-t)_+}_{=0} \right) \cdot x + \left((1-t)^2 - 2 \underbrace{(0-t)_+}_{=0} \right) x^2 \\
&= (x-t)_+^2 + 0 + ((1-t)^2 - 0) x^2 \\
&= (x-t)_+^2 + x^2 (1-t)^2 > 0
\end{aligned}$$

$$0 \leq x \leq t \leq 1 \Rightarrow (x-t)_+ = 0$$

$$\hookrightarrow 0^2 + x^2 (1-t)^2 > 0$$

$$0 \leq t \leq x \leq 1 \Rightarrow (x-t)_+ = x-t \left. \vphantom{\begin{matrix} \hookrightarrow \\ \hookrightarrow \end{matrix}} \right\} \Rightarrow$$

$$\hookrightarrow (x-t)^2 + x^2 (1-t)^2 > 0$$

$\Rightarrow K$ has constant sign on $[0, 1] \Rightarrow$

corollary Peano $\Rightarrow Rf = \frac{1}{3!} f^{(3)}(\xi) \cdot Re_3$

$$R_2 f(x) = \int_0^1 \frac{1}{3!} f^{(3)}(\xi) \cdot Re_3 = \frac{1}{6} f^{(3)}(\xi) \int_0^1 Re_3$$

$$= \frac{1}{6} f^{(3)}(\xi) \cdot R_2 e_3(x) = \frac{1}{6} f^{(3)}(\xi) \cdot \begin{pmatrix} -\frac{1}{12} \end{pmatrix}$$

$$\int_0^2 x^2$$

$$= \frac{1}{6} f'(x) \cdot R_2 e_3(x) = \frac{1}{6} f'(x) \cdot \begin{pmatrix} -12 \\ 1 \end{pmatrix}$$

$$= -\frac{1}{6} f^{(3)}(x)$$

2) (2.5 points) Find the Cholesky decomposition of the Hilbert matrix

$$H_3 = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix} \sim \left[\begin{array}{cc|c} \sqrt{1} & & \\ 1/2 & & \\ 1/3 & & \end{array} \right]$$

$$A' = A - w w^T / a_{11} = \begin{bmatrix} 1/3 & 1/4 \\ 1/4 & 1/5 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/2 & 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 & 1/4 \\ 1/4 & 1/5 \end{bmatrix} - \begin{bmatrix} 1/6 & 1/6 \\ 1/6 & 1/9 \end{bmatrix} = \begin{bmatrix} 1/12 & 1/6 \\ 1/6 & 4/45 \end{bmatrix}$$

$$A \sim \left[\begin{array}{c|cc} 1 & & \\ 1/2 & 1/12 & \\ 1/3 & 1/6 & 4/45 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & & \\ 1/2 & 1/\sqrt{12} & \\ 1/3 & \underbrace{1/6 \cdot \frac{\sqrt{12}}{1}}_{\frac{\sqrt{12}}{12}} & \end{array} \right]$$

$$A' - w w^T / a_{21} = \begin{bmatrix} 4/45 \end{bmatrix} - \begin{bmatrix} 1/12 \end{bmatrix} \cdot \begin{bmatrix} 1/12 \end{bmatrix} / 1/12$$

$$= \begin{bmatrix} 1/45 \end{bmatrix} - \left(1/144 \cdot \frac{12}{1} \right) = \begin{bmatrix} 4/45 \end{bmatrix} - \begin{bmatrix} \frac{1}{12} \end{bmatrix} = \begin{bmatrix} \frac{1}{45} \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & & \\ 1/2 & 1/\sqrt{12} & \\ 1/3 & \frac{1}{\sqrt{12}} & 1/80 \end{bmatrix} \Rightarrow R^T \sim \begin{bmatrix} 1 & 0 \\ 1/2 & 1/\sqrt{12} \\ 1/3 & 1/\sqrt{12} \end{bmatrix}$$

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$$\begin{array}{r} 15) \\ \frac{+}{5} - \frac{1}{12} \\ \hline 6 - 15 = \frac{1}{180} \end{array}$$

$$\begin{array}{r} 0 \\ 0 \\ \frac{1}{180} \end{array} \Bigg]$$

$$R \sim \begin{bmatrix} 1 & 1/2 & 1/3 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{80} \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix} \begin{matrix} \nearrow [a_{12} \ a_{13}] \\ \searrow \begin{bmatrix} a_{21} \\ a_{31} \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & w^* \\ v/a_{11} & A' - v \cdot w^*/a_{11} \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 2 \end{bmatrix} / 2 =$$

$$= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 2 \\ 4 & 2 \end{bmatrix} / 2 =$$

$$\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} / (-1) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 1/2 & 1 & 1 \end{bmatrix}$$

$$\text{Cholesky} \Rightarrow A = \begin{bmatrix} a_{11} & w^T \\ w & A' \end{bmatrix} \quad A = R^T \cdot R$$

Cholesky $\Rightarrow$

$$A = \begin{bmatrix} a_{11} & w \\ w & A' \end{bmatrix}$$

$$A = R \cdot R^T$$

2) (2.5 points) Find the Cholesky decomposition of the Hilbert matrix

$$H_3 = \begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{a_{11}} & \\ w/\sqrt{a_{11}} & A' - ww^T/a_{11} \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ 1/2 & 1/3 & \\ 1/3 & 1/4 & 1/5 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & & \\ 1/2 & 1/3 & \\ 1/3 & 1/4 & 1/5 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} \sqrt{1} & & \\ (1/2)/\sqrt{1} & 1/3 & \\ (1/3)/\sqrt{1} & 1/4 & 1/5 \end{bmatrix}$$

$$\begin{bmatrix} 1/3 \\ 1/4 & 1/5 \end{bmatrix}$$

$$- \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1/2 & 1/3 \end{bmatrix} / 1 =$$

$$= \begin{bmatrix} 1/3 \\ 1/4 & 1/5 \end{bmatrix} - \begin{bmatrix} 1/6 & 1/9 \end{bmatrix} = \begin{bmatrix} 1/12 & \\ 1/12 & 4/45 \end{bmatrix}$$

$$\begin{bmatrix} 1/12 & \\ 1/12 & 4/45 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1/\sqrt{12} & \\ 1/\sqrt{12} & \end{bmatrix}$$

$$\begin{bmatrix} 4/45 \end{bmatrix} - \begin{bmatrix} 1/12 \end{bmatrix} \cdot \begin{bmatrix} 1/12 \end{bmatrix} / \frac{1}{12} = \frac{4}{45} - \frac{1}{12} = \frac{1}{180} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} 1 & & \\ 1/2 & 1/\sqrt{12} & \\ 1/3 & 1/\sqrt{12} & 1/\sqrt{180} \end{bmatrix} \Rightarrow$$

$$R^T =$$

$$\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2\sqrt{3} \\ 1/3 & 1/2\sqrt{3} \end{bmatrix}$$

$$R =$$

$$\begin{bmatrix} 1 & 1/2 \\ 0 & 1/2\sqrt{3} \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1/\sqrt{12} \\ 1/\sqrt{12} \quad \boxed{\frac{1}{180}} \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} 1/\sqrt{12} \\ 1/\sqrt{12} \quad 1/\sqrt{180} \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1/6\sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} 1/3 \\ 1/2\sqrt{3} \\ 1/6\sqrt{5} \end{bmatrix}$$

